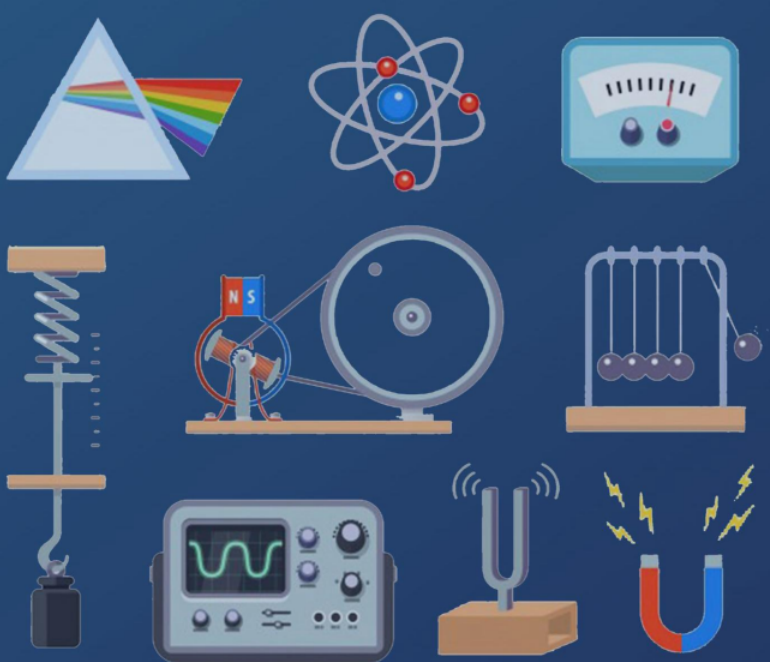
PHYSICS

Elasticity



SHEET

eSaral.com

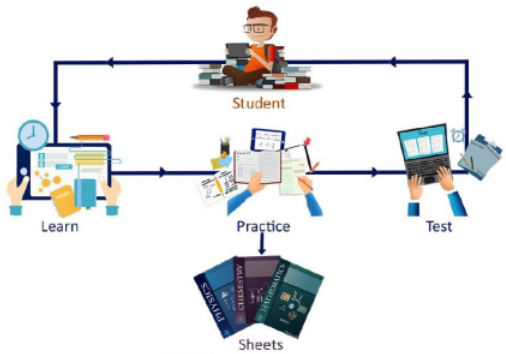
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Stress and Strain	1	1,4,6,7	1	1
2	Hooke's Law and Elastic Moduli	2-7	2,5	2-5	2-4
3	Stress-Strain Curve and Elastic Potential Energy	8,9	3		5,6

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Stress and Strain

Q1. One end of uniform wire of length L and of weight W is attached rigidly to a point in the roof and a weight W_1 is suspended from its lower end. If S is the area of cross-section of the wire, the stress in the wire at a height (L/4) from its lower end is

(A) $\frac{W_1}{S}$ (B) $\frac{\left[W_1+\frac{W}{4}\right]}{S}$

(C) $\frac{\left[W_1+\frac{3W}{4}\right]}{S}$ (D) $\frac{W_1+W}{4}$

Hooke's Law and Elastic Moduli

Q2. The bulk modulus of copper is 1.4×10^{11} Pa and the coefficient of linear expansion is 1.7×10^{-5} $(C°)^{-1}$. What hydrostatic pressure is necessary to prevent a copper block from expanding when its temperature is increased from 20°C to 30°C?

(A) 6.0×10^5 Pa (B) 7.1×10^7 Pa

(C) 5.2×10^6 Pa (D) 40 atm

Q3. The pressure of a medium is changed from 1.01×10^5 Pa to 1.165×10^5 Pa and change in volume is 10% keeping temperature constant. The bulk modulus of the medium is :–

(A) 204.8×10^5 Pa

(B) 102.4×10^5 Pa

(C) 51.2×10^5 Pa

(D) 1.55×10^5 Pa

Q4. A highly rigid cubical block A of small mass M and side L is fixed rigidly onto another cubical block B of same dimensions and of low modulus of rigidly η such that lower face of A completely covers the upper face of B. The lower face of B is rigidly held on a horizontal surface. A small force F is applied perpendicular to one of the side face of A. After the force is withdrawn, block A executes small oscillations, the time period of which is given by-

(A) $2\pi\sqrt{M\eta L}$ (B) $2\pi\sqrt{\frac{M}{\eta L}}$

(C) $2\pi\sqrt{\frac{ML}{\eta}}$ (D) $\sqrt{\frac{M\eta}{L}}$

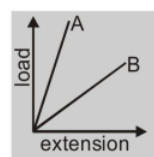
Q5. The dimensions of two wires A and B are the same. But their materials are different. Their load-extension graphs are shown. If Y_A and Y_B are the values of Young's modulus of elasticity of A and B respectively then -

(A) $Y_A > Y_B$

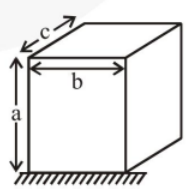
(B) $Y_A < Y_B$

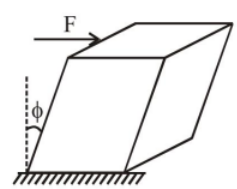
(C) $Y_A = Y_B$

(D) $Y_B = 2Y_A$



Q6. A cuboidal block of sides a, b & c is fixed on ground. The top is pushed by a horizontal force F as shown. The angle ϕ by which the block deforms is : (η is modulus of rigidity)





(A) $\frac{F}{ab\eta}$ (B) $\frac{F}{ac\eta}$

(C) $\frac{F}{bc\eta}$ (D) $\frac{F}{\sqrt{b^2+c^2}\,\eta}$

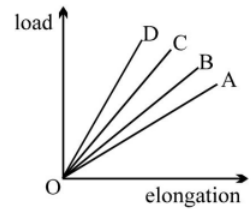
Q7. A force F is needed to break a copper wire having radius R. The force needed to break a copper wire of radius 2R will be :-

(A) $\frac{F}{2}$ (B) 2F

(C) 4F (D) $\frac{F}{4}$

Stress-Strain Curve and Elastic Potential Energy

Q8. The load versus strain graph for four wires of the same material is shown in the figure. The thickest wire is represented by the line:-



(A) OB (B) OA (C) OD (D) OC

Q9. Two wires of the same material and length but diameters in the ratio 1 : 2 are stretched by the same force. The potential energy per unit volume for the two wires when stretched will be in the ratio.

(A) 16 : 1 (B) 4 : 1

(C) 2 : 1 (D) 1 : 1

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. A wire is suspended from the ceiling and stretched under the action of a weight *F* suspended from its other end. The force exerted by the ceiling on it is equal and opposite to the weight.

(A) Tensile stress at any cross section *A* of the wire is *F/A*.

(B) Tensile stress at any cross section is zero.

(C) Tensile stress at any cross section *A* of the wire is *2F/A*.

(D) Tension at any cross section A of the wire is *F*.

Q2. The temperature of a wire is doubled. The Young's modulus of elasticity

(A) will also double.

(B) will become four times.

(C) will remain same.

(D) will decrease.

Q3. A wire fixed at the upper end stretches by length ℓ by applying a force F. The work done in stretching is-

(A) $\frac{F}{2\ell}$ (B) $F\ell$

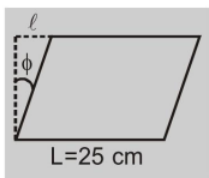
(C) $2F\ell$ (D) $\frac{F\ell}{2}$

NUMERICAL TYPE

The answer to each question is a ***NUMERICAL VALUE****. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

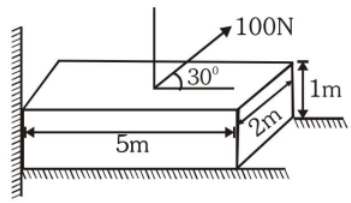
Q4. The breaking stress of steel is 7.9×10^8 N/m² and density is 7.9×10^3 kg/m³. What should be the maximum length so that it may not break with its own weight is .....km.

Q5. Two parallel and opposite forces, each 4000 N, are applied tangentially to the upper and lower faces of a cubical metal block of side 25 cm. If the shear modulus for the metal is 8×10^{10} Pa, then the displacement of the upper surface relative to the lower surface will be ..... m.



SUBJECTIVE TYPE

Q6. Find out longitudinal stress and tangential stress on a fixed block.



Q7. A human bone is subjected to a compressive force of 5.0×10^5 N. The bone is 25 cm long and has an approximate cross sectional area 4.0 cm². If the ultimate compressive strength of the bone is 1.70×10^8 N/m², will the bone be compressed or it will break under this force ?

Exercise - 2 *Brain Booster*

SINGLE CORRECT TYPE

Each question has FOUR options for correct answer. ***ONLY ONE*** *of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. Steel ruptures when a shear of 3.5×10^8 Nm^{-2} is applied. The force needed to punch a 1 cm diameter hole in a steel sheet 0.3 cm thick is nearly :-

(A) 3.3×10^4 N (B) 1.4×10^4 N

(C) 1.1×10^4 N (D) 2.7×10^4 N

Q2. A solid sphere of radius R made of material of bulk modulus K is surrounded by a liquid in a cylindrical container. A massless piston of area A floats on the surface of the liquid. When a mass m is placed on the piston to compress the liquid, the fractional change in the radius of the sphere $\Delta R/R$ is

(A) mg/AK (B) mg/3AK

(C) mg/A (D) mg/3AR

Q3. A steel wire 1.5 m long and of radius 1 mm is attached with a load 3 kg at one end the other end of the wire is fixed, it is whirled in a vertical circle with a frequency 2Hz. Find the elongation of the wire when the weight is at the lowest position–

($Y = 2 \times 10^{11}$ N/m² and $g = 10$ m/s²)

(A) 1.77×10^{-3} m

(B) 7.17×10^{-3} m

(C) 3.17×10^{-7} m

(D) 1.37×10^{-7} m

Q4. The Young's modulus of a rubber string 8 cm long and density 1.5 kg/m³ is 5×10^8 N/m², is suspended on the ceiling in a room. The increase in length due to its own weight will be :

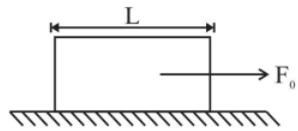
(A) 9.6×10^{-5} m

(B) 9.6×10^{-11} m

(C) 9.6×10^{-3} m

(D) 9.6 m

Q5. A constant force F_0 is applied on a uniform elastic string placed over a smooth horizontal surface as shown in figure. Young's modulus of string is Y and area of cross - section is S.The strain produced in the string in the direction of force is -



(A) $\frac{F_0 Y}{S}$ (B) $\frac{F_0}{SY}$ (C) $\frac{F_0}{2SY}$ (D) $\frac{F_0 Y}{2S}$

Exercise - 2A *Brain Booster*

ONE OR MORE THAN ONE CORRECT TYPE

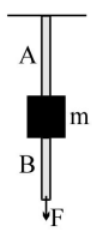
*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. A copper and a steel wire of the same diameter are connected end to end. A deforming force F is applied to this composite wire which causes a total elongation of 1cm. The two wires will have

(A) the same stress. (B) different stress.

(C) the same strain. (D) different strain.

Q2. The wires A and B shown in the figure are made of the same material and have radii r_A and r_B respectively. The block between them has a mass m. When the force F is mg/3, one of the wires breaks.



(A) A breaks if $r_A = r_B$

(B) A breaks if $r_A < 2r_B$

(C) Either A or B may break if $r_A = 2r_B$

(D) The lengths of A and B must be known to predict which wire will break

Q3. Overall changes in volume and radii of a uniform cylindrical steel wire are 0.2% and 0.002% respectively when subjected to some suitable force. Longitudinal tensile stress acting on the wire is :-

($Y = 2.0 \times 10^{11}$ Nm^{-2})

(A) 3.2×10^9 Nm^{-2} (B) 3.2×10^7 Nm^{-2}

(C) 3.6×10^9 Nm^{-2} (D) 3.9×10^8 Nm^{-2}

Q4. A cylindrical wire of radius 1 mm, length 1 m, Young's modulus = 2×10^{11} N/m^2, poisson's ratio $\mu = \pi/10$ is stretched by a force of 100 N. Its radius will become :

(A) 0.99998 mm (B) 0.99999 mm

(C) 0.99997 mm (D) 0.99995 mm

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q5. A wire suspended vertically from one of its ends is stretched by attaching a weight of 200 N to the lower ends. The weight stretches the wire by 1mm. Then the elastic energy stored in the wire is-

Q6. Young modulus of elasticity of brass is 10^{11} N/m^2. On pressing the rod of length 0.1 m and cross–sectional area 1 cm^2 made from brass with a force of 10 N along its length, the increase in its energy will be ..... J.

Answer Key

Ex-1								
1	2	3	4	5	6	7	8	9
B	B	D	B	A	C	C	C	A

Ex-1A

1. A,D **2.** D **3.** D

4. 10 **5.** 2×10^{-7}

6. $5\sqrt{3}$ N/m^2 , 5 N/m^2 **7.** Bone will break

Ex-2				
1	2	3	4	5
A	B	A	B	C

Ex-2A

1. A,D **2.** A,B,C **3.** D

4. D **5.** 0.1 J **6.** 5×10^{-7}

-: Important Notes & Formulas :-